Antimicrobial Stewardship: Antibiotic Utilization Analysis

Using Excel, Power Query, Power Pivot, DAX and PivotTables to explore antibiotic utilization and WHO AWaRe classification

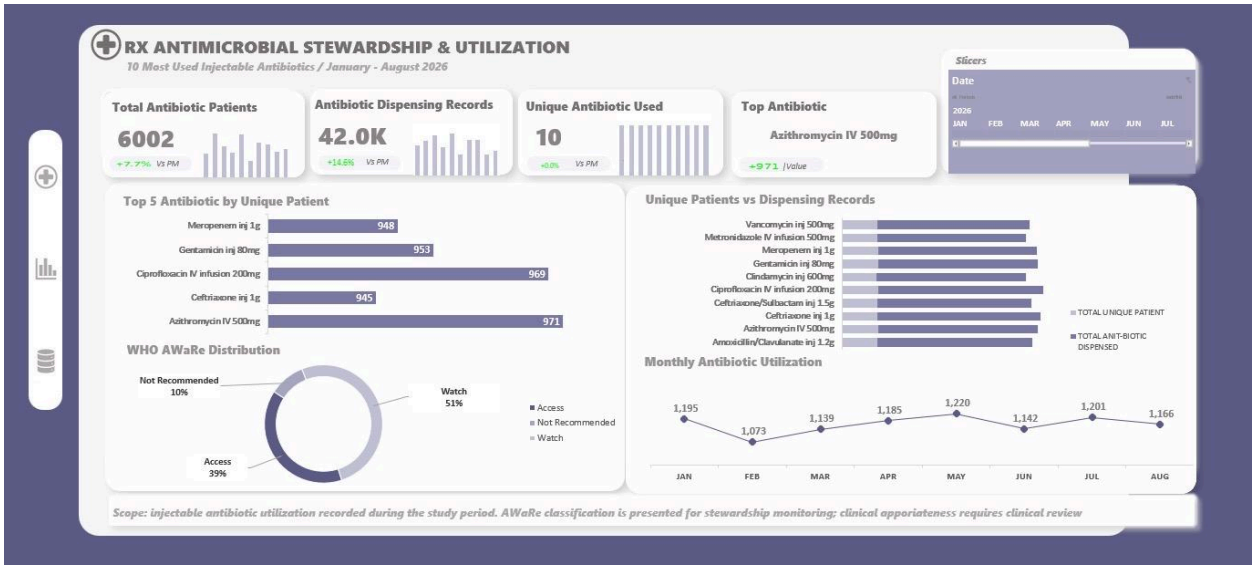
1. Project Overview

Antimicrobial resistance is an increasing global healthcare concern, making responsible antibiotic use and antimicrobial stewardship increasingly important.

This project was inspired by a hospital pharmacy antimicrobial stewardship activity focused on understanding antibiotic utilization within a private hospital setting.

The analytical objective was to explore **which injectable antibiotics were being used most frequently, how many unique patients received them, how utilization changed over time, and how the antibiotics were distributed across the WHO AWaRe classification framework.**

Because real hospital and patient data are confidential, a **synthetic dataset containing 10,000+ antibiotic dispensing records** was created for this portfolio project.



2. Business / Healthcare Problem

Antibiotic dispensing records can provide valuable information, but raw transaction data does not immediately reveal utilization patterns.

The project aimed to answer:

- Which antibiotics had the highest utilization?
- How many unique patients received antibiotics?
- What was the total quantity dispensed?
- How did antibiotic utilization change monthly?
- Which antibiotics appeared most frequently among patients?
- How were the antibiotics distributed across WHO AWaRe classifications?
- Could the resulting analysis provide a useful starting point for antimicrobial stewardship discussions?

The objective was **not to determine whether individual prescriptions were clinically appropriate**, but to make utilization patterns visible for further review.

3. Dataset

The synthetic dataset represented antibiotic dispensing activity from **January to August**.

Dataset characteristics

- **10,000+** synthetic dispensing records
- **6,002** unique patients
- **10** unique antibiotics
- Injectable antibiotics
- Patient-level identifiers
- Antibiotic identifiers
- Quantity dispensed
- Transaction information
- Dates
- WHO AWaRe classification mapping

No real patient information or confidential hospital records were used in the published analysis.

Anti-Biotic	Brand	Type	Number	Customer	Customer_pid	Customer_name	Quantity	Reference	Price	WHO	Date
Metronidazole IV infusion 500mg	Flagyl IV	Sales Invoice	449000	INS-45-B2	PID3286	NURSE BALOGUN	10	455100	4500	Q222	09/07/2026
Clindamycin inj 600mg	Dalacin C IV	Sales Invoice	449001	INS-85-D1	PID1488	BLESSING EZE	2	455101	14000	Q999	10/05/2026
Azithromycin IV 500mg	Zithromax IV	Sales Invoice	449002	INS-81-B5	PID7873	SARAH BELLO	5	455102	19500	Q1010	13/03/2026
Ceftriaxone inj 1g	Rocephine	Sales Invoice	449003	INS-99-D3	PID5552	AMINU EZE	3	455103	15000	Q111	27/01/2026
Metronidazole IV infusion 500mg	Flagyl IV	Sales Invoice	449004	22981052	PID6881	PATIENT MUSA	5	455104	4500	Q222	09/03/2026
Ceftriaxone inj 1g	Rocephine	Sales Invoice	449005	81971316	PID3045	FATIMA BALOGUN	5	455105	15000	Q111	17/03/2026
Azithromycin IV 500mg	Zithromax IV	Sales Invoice	449006	67490893	PID4150	NURSE BALOGUN	1	455106	19500	Q1010	19/06/2026
Amoxicillin/Clavulanate inj 1.2g	Augmentin IV	Sales Invoice	449007	20709497	PID4814	PATIENT BALOGUN	4	455107	18000	Q444	13/03/2026
Vancomycin inj 500mg	Vancocin	Sales Invoice	449008	31831063	PID7065	EMEKA EZE	10	455108	28000	Q888	10/03/2026
Metronidazole IV infusion 500mg	Flagyl IV	Sales Invoice	449009	INS-78-B2	PID8573	FATIMA OKON	10	455109	4500	Q222	26/06/2026
Clindamycin inj 600mg	Dalacin C IV	Sales Invoice	449010	INS-97-C1	PID4752	PATIENT OBI	3	455110	14000	Q999	13/04/2026
Gentamicin inj 80mg	Gentrisone	Sales Invoice	449011	INS-37-C2	PID9179	FATIMA UMAR	4	455111	2500	Q555	06/02/2026
Gentamicin inj 80mg	Gentrisone	Sales Invoice	449012	INS-41-C5	PID8019	IBRAHIM ADEDAYO	3	455112	2500	Q555	26/02/2026
Ciprofloxacin IV infusion 200mg	Ciprotab IV	Sales Invoice	449013	22201654	PID1771	PATIENT BALOGUN	2	455113	6000	Q333	10/06/2026
Ciprofloxacin IV infusion 200mg	Ciprotab IV	Sales Invoice	449014	90048665	PID2040	FATIMA ADEDAYO	5	455114	6000	Q333	30/04/2026
Clindamycin inj 600mg	Dalacin C IV	Sales Invoice	449015	84252722	PID1188	YETUNDE BALOGUN	10	455115	14000	Q999	15/08/2026
Clindamycin inj 600mg	Dalacin C IV	Sales Invoice	449016	96028436	PID6573	BLESSING OKON	4	455116	14000	Q999	10/02/2026
Vancomycin inj 500mg	Vancocin	Sales Invoice	449017	INS-43-B5	PID2743	PATIENT UMAR	3	455117	28000	Q888	04/08/2026
Clindamycin inj 600mg	Dalacin C IV	Sales Invoice	449018	INS-29-C2	PID9837	DR CHUKWU	1	455118	14000	Q999	03/06/2026
Meropenem inj 1g	Meronem	Sales Invoice	449019	12614124	PID2832	EMEKA OKON	2	455119	35000	Q666	15/01/2026
Amoxicillin/Clavulanate inj 1.2g	Augmentin IV	Sales Invoice	449020	INS-20-D1	PID9727	DR SMITH	2	455120	18000	Q444	18/06/2026
Vancomycin inj 500mg	Vancocin	Sales Invoice	449021	INS-43-D2	PID9835	DR EZE	10	455121	28000	Q888	21/03/2026
Ceftriaxone/Sulbactam inj 1.5g	Tandak	Sales Invoice	449022	68800797	PID9479	TUNDE BALOGUN	2	455122	42000	Q777	27/02/2026
Metronidazole IV infusion 500mg	Flagyl IV	Sales Invoice	449023	12823170	PID4770	IBRAHIM EZE	1	455123	4500	Q222	19/01/2026
Ceftriaxone inj 1g	Rocephine	Sales Invoice	449024	INS-18-A3	PID2160	CHINELO EZE	3	455124	15000	Q111	21/06/2026
Vancomycin inj 500mg	Vancocin	Sales Invoice	449025	INS-79-B5	PID8744	SARAH BELLO	4	455125	28000	Q888	18/02/2026
Metronidazole IV infusion 500mg	Flagyl IV	Sales Invoice	449026	INS-94-D3	PID7939	FATIMA BELLO	10	455126	4500	Q222	14/01/2026
Metronidazole IV infusion 500mg	Flagyl IV	Sales Invoice	449027	INS-61-C1	PID5073	SARAH EZE	5	455127	4500	Q222	25/04/2026
Ciprofloxacin IV infusion 200mg	Ciprotab IV	Sales Invoice	449028	34627347	PID5563	TUNDE EZE	1	455128	6000	Q333	24/04/2026
Clindamycin inj 600mg	Dalacin C IV	Sales Invoice	449029	INS-16-A1	PID4872	AMINU ADEDAYO	4	455129	14000	Q999	04/05/2026
Amoxicillin/Clavulanate inj 1.2g	Augmentin IV	Sales Invoice	449030	17869910	PID3697	FATIMA OBI	4	455130	18000	Q444	09/03/2026
Vancomycin inj 500mg	Vancocin	Sales Invoice	449031	66775103	PID8973	AMINU EZE	3	455131	28000	Q888	25/02/2026
Ceftriaxone inj 1g	Rocephine	Sales Invoice	449032	INS-50-A1	PID8811	CHINELO CHUKWU	2	455132	15000	Q111	15/01/2026
Clindamycin inj 600mg	Dalacin C IV	Sales Invoice	449033	INS-33-A5	PID2113	YETUNDE EZE	4	455133	14000	Q999	31/01/2026
Azithromycin IV 500mg	Zithromax IV	Sales Invoice	449034	INS-84-A5	PID2343	FATIMA UMAR	5	455134	19500	Q1010	25/05/2026
Clindamycin inj 600mg	Dalacin C IV	Sales Invoice	449035	44999379	PID4346	YETUNDE MUSA	2	455135	14000	Q999	09/03/2026
Ceftriaxone/Sulbactam inj 1.5g	Tandak	Sales Invoice	449036	INS-95-C4	PID6180	DR BALOGUN	1	455136	42000	Q777	28/04/2026
Azithromycin IV 500mg	Zithromax IV	Sales Invoice	449037	INS-19-B5	PID5345	AMINU MUSA	1	455137	19500	Q1010	14/08/2026
Amoxicillin/Clavulanate inj 1.2g	Augmentin IV	Sales Invoice	449038	48250360	PID3584	TUNDE CHUKWU	10	455138	18000	Q444	19/03/2026
Azithromycin IV 500mg	Zithromax IV	Sales Invoice	449039	INS-95-C1	PID3200	GRACE BALOGUN	1	455139	19500	Q1010	10/07/2026

4. Data Preparation

The first stage of the project was data cleaning and preparation.

I reviewed the available fields and removed information that was not relevant to the utilization questions.

Cleaning steps included:

- Removing irrelevant columns
- Removing unnecessary blank/null records
- Removing duplicates
- Excluding stock-adjustment transactions from utilization analysis
- Renaming columns to accurately represent their meaning
- Creating unique antibiotic identifiers
- Mapping antibiotics to WHO AWaRe classifications
- Preparing the dataset for Power Query and Power Pivot

Why stock adjustments were removed

Stock adjustments represent inventory movement rather than medication dispensed to a patient.

Including these records in a patient utilization analysis could therefore inflate antibiotic utilization.

This was an important data-cleaning decision because **what is included in the dataset determines what the analysis represents.**

5. Power Query Transformation

Power Query was used to prepare the dataset for analysis.

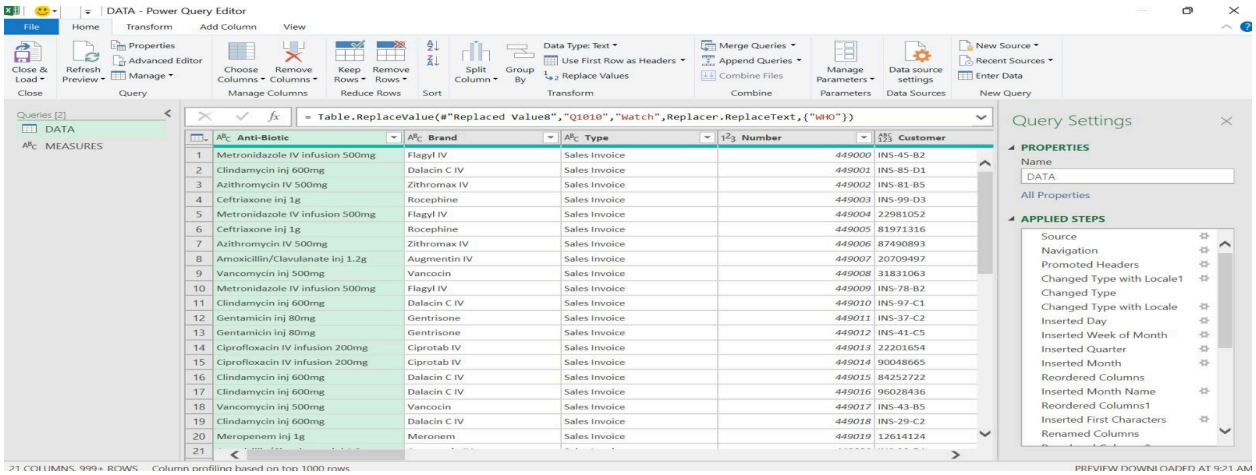
Key transformations included:

- Setting appropriate data types
- Correcting date interpretation
- Using **Change Type with Locale → English (United Kingdom)** to correctly interpret the dates
- Creating additional date fields
- Mapping antibiotic identifiers to WHO AWaRe classifications

The date structure included:

- Date
- Day
- Month Number
- Month Name
- Short Month Name
- Quarter
- Year

These fields supported the monthly utilization analysis and the dashboard's date timeline/slicer.



6. WHO AWaRe Classification Mapping

Each antibiotic was assigned a unique identifier.

For example:

Q111 → Ceftriaxone injection

The identifiers were then used in Power Query to map the antibiotics to their corresponding WHO AWaRe classifications.

This allowed the classifications to be incorporated into the analytical dataset and used to calculate the distribution displayed in the dashboard.

The analysis included:

- Access
- Watch
- Not Recommended

The classification was used for **descriptive analysis**, not as a clinical determination of whether an individual prescription was appropriate.

7. Power Pivot and DAX

Power Pivot was used to support the analytical model within Excel.

Although I had previous exposure to DAX, this was my first time implementing DAX within **Excel Power Pivot**.

Measures were created to calculate key indicators such as:

Total Unique Patients

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Total Unique Patients =
DISTINCTCOUNT(Data[Customer_PID])
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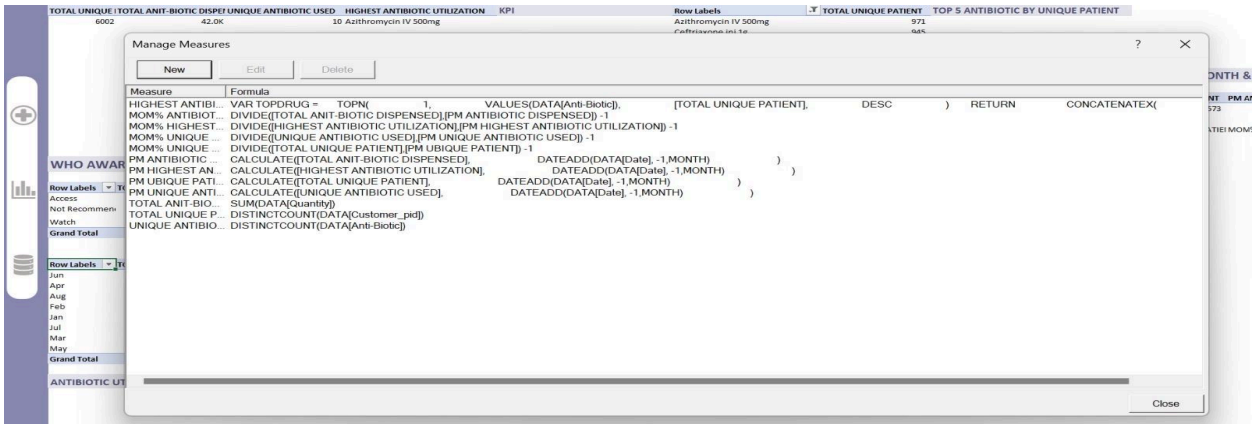
Total Antibiotic Quantity Dispensed

Total Antibiotic Dispensed =
SUM(Data[Quantity])

Additional measures supported:

- Unique antibiotics
- Highest antibiotic utilization
- Previous-month values
- Month-over-month percentage changes
- Supporting KPI trends

Previous-month calculations used CALCULATE and DATEADD, while DIVIDE was used to calculate percentage changes.



8. PivotTable Analysis

PivotTables were used to explore the dataset before and during dashboard development.

The analysis included:

Top 5 Antibiotics by Unique Patients

Identifying which antibiotics were received by the largest number of unique patients.

Monthly Antibiotic Utilization

Showing changes in unique patient utilization from January through August.

WHO AWaRe Distribution

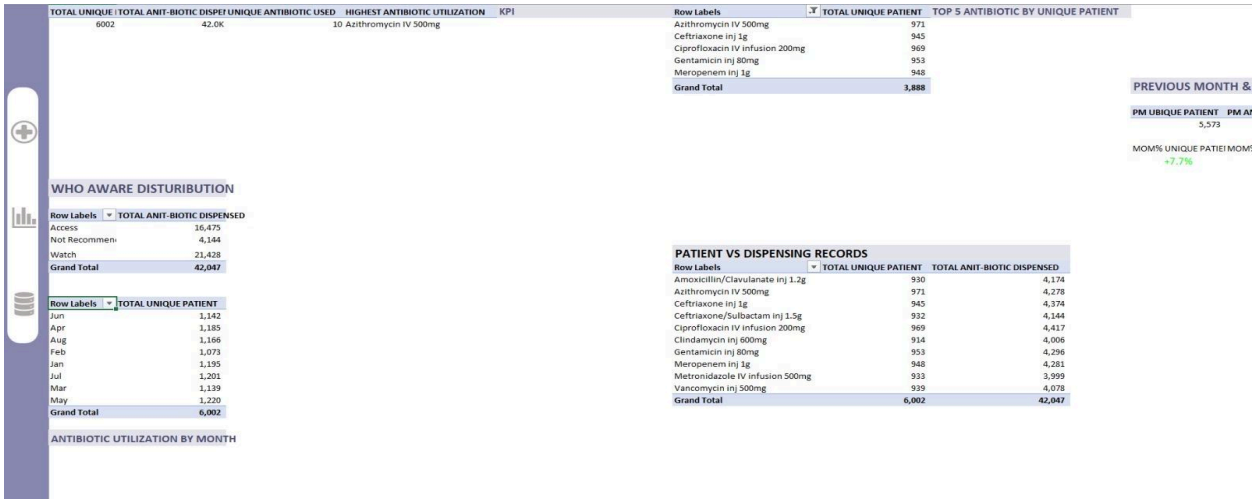
Showing the proportion of antibiotic utilization across the classifications.

Patient vs. Dispensing Quantity

Comparing unique patients with the overall quantity of antibiotics dispensed.

KPI and Supporting Trends

Supporting the metrics and trend indicators displayed in the final dashboard.



9. Key Findings

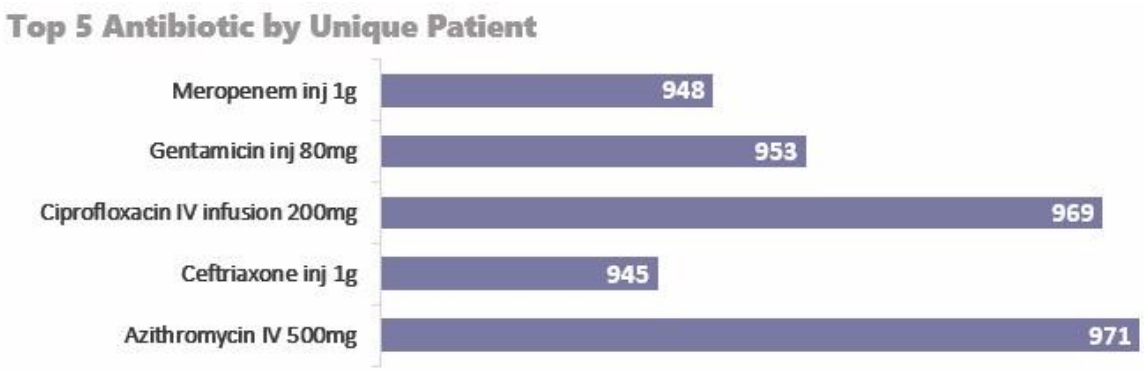
The final dashboard produced several key indicators from the synthetic dataset.

KPI	Result
Total Unique Patients	**6,002**
Antibiotic Quantity Dispensed	**~42,000**

Unique Antibiotics	**10**
Highest Antibiotic Utilization	**Azithromycin IV 500mg**
Highest Unique Patient Count	**971**
Highest Monthly Utilization	**May – 1,220 unique patients**

Top antibiotics by unique patient count

Antibiotic	Unique Patients
Azithromycin IV 500mg	971
Ciprofloxacin IV	969
Gentamicin Injection 80mg	953
Ciprofloxacin IV Infusion 200mg	969
Meropenem Injection 1g	948



10. WHO AWaRe Distribution

The dashboard showed the following distribution:

AWaRe Classification	Share
Watch	**51%**
Access	**39%**
Not Recommended	**10%**



This provides another perspective beyond simply looking at antibiotic volume.

However, the distribution **does not by itself indicate inappropriate prescribing**. Clinical indication, diagnosis, treatment duration, patient condition and other clinical factors would be required for a proper stewardship assessment.

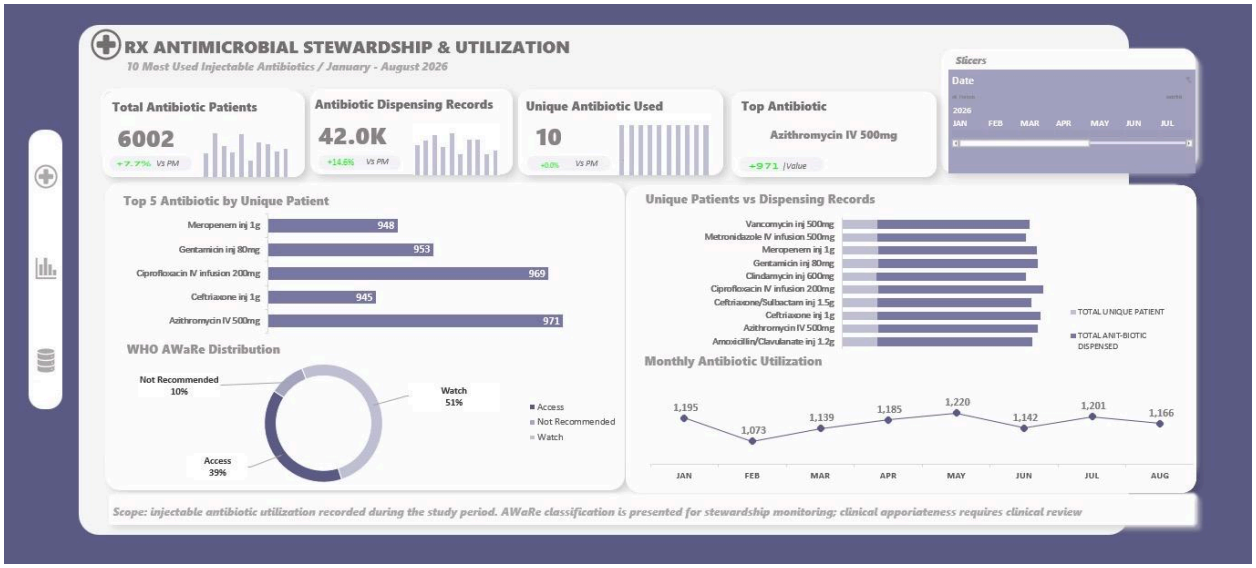
11. Dashboard

The final Excel dashboard brings the analysis together into a single interactive view.

It includes:

- Total unique patients
- Total antibiotic quantity dispensed
- Unique antibiotics
- Highest-utilized antibiotic
- Top 5 antibiotics by unique patients
- Monthly antibiotic utilization
- Patient vs. dispensing quantity
- WHO AWaRe distribution
- KPI trend indicators
- Date timeline/slicer

The dashboard transforms thousands of individual records into a format that allows users to quickly identify utilization patterns and areas that may require further investigation.



12. Key Insight

One of my biggest takeaways was seeing the difference between **dispensing quantity and unique patient utilization**.

The dataset contained approximately **42,000 units of antibiotics dispensed**, while the dashboard reported **6,002 unique patients**.

This reinforced an important analytical principle:

A dispensing record, a quantity dispensed and a patient are three different things.

Simply counting rows would not have provided the same insight.

The project also pushed me to research antimicrobial resistance more deeply and think about how healthcare data can support conversations around antibiotic utilization and stewardship.

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13. Limitations

This project has several limitations.

- The dataset is synthetic and does not represent actual hospital utilization.
- It does not contain clinical diagnoses or treatment indications.

- It does not assess individual prescription appropriateness.
- The analysis is descriptive rather than a complete antimicrobial stewardship audit.
- The WHO AWaRe distribution should not be interpreted alone as evidence of inappropriate antibiotic use.

A real-world stewardship program would require additional clinical and patient-level information.

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14. Recommendations for Further Analysis

If this were developed into a real hospital analytical solution, future analysis could include:

- Antibiotic utilization by diagnosis
- Duration of therapy
- Antibiotic switching/de-escalation
- Culture and sensitivity results
- Prescribing department
- Prescriber-level trends
- Antibiotic consumption by ward
- Resistance patterns
- Antibiotic expenditure
- Stock availability and stockouts
- Appropriate versus inappropriate utilization
- Trends in resistant organisms

This would move the dashboard from **utilization monitoring** toward a more comprehensive antimicrobial stewardship analytics system.

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15. Conclusion

This project demonstrated how pharmaceutical dispensing data can be transformed into meaningful healthcare intelligence using familiar analytical tools.

The workflow moved from:

Raw Data → Cleaning → Power Query → Power Pivot/DAX → PivotTables → Dashboard → Healthcare Insights

For me, the project was particularly valuable because it combined my background as a **Pharmacy Technician** with my developing skills in **healthcare data analytics**.

The goal was not simply to create a dashboard.

It was to demonstrate how data can help healthcare teams **see utilization patterns, ask better questions and identify areas that may require further clinical review.**

Project Resources

Interactive Excel Dashboard:
Explore the Excel workbook, including the interactive dashboard, PivotTables, and analytical measures.

Click here to view the Excel workbook

Replace this text with a clickable link to your shared Excel file.

Portfolio:
Explore my other data analytics projects and professional work.

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